METRIC

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DETAIL SPECIFICATION

PROPYLENE GLYCOL DINITRATE

This specification is approved for use by the Naval Surface Warfare Center Indian Head Division (NSWC IHD) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of propylene glycol dinitrate (PGDN) for use in the manufacture of propellants.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-59282 - Chemicals, Analytical; General Specification for

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-286 - Propellants, Solid: Sampling, Examination and Testing

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 9135

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NAVAL SURFACE WARFARE CENTER INDIAN HEAD DIVISION (NSWC IHD) PUBLICATIONS

LP-AAL-23002 - 1,2-Propanediol Testing for PGDN Production

LP-AAL-21008 - Propylene Glycol Characterization for the Production of Propylene Glycol Dinitrate

(Copies of these documents are available from Naval Surface Warfare Center Indian Head Division, Code E16, 3703 Benson Road, Indian Head, MD 20640-5116.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS411-1 - Hazardous Materials Target List

(Copies of this document are available online at www.aia-aerospace.org.)

ASTM INTERNATIONAL

ASTM D1209 - Standard Test Method for Color of Clear Liquids (Platinum-Cobalt Scale)

ASTM D1218 - Standard Test Method for Refractive Index and Refractive Dispersion of Hydrocarbon Liquids

ASTM E203 - Standard Test Method for Water Using Volumetric Karl Fischer Titration

(Copies of these documents are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. The PGDN shall be a product made from 1,2-propanediol (herein referred to as propylene glycol) that meets or exceeds the requirements of table I (see 4.2.1 and 6.7).

TABLE I. Minimum requirements for propylene glycol.

Test Description	Min	Max	Unit of Measure	Test Method
Color	-	10	Pt-Co Scale	ASTM D1209
Specific gravity at 25/25 °C	1.0352	1.0364	-	LP-AAL-23002
Acidity (as acetic acid)	-	0.002	Weight %	LP-AAL-23002
Refractive index	1.431	1.433	n_D25 °C	ASTM D1218
Purity (on dry basis)	99.8	-	Weight %	LP-AAL-21008
Carbonyl	-	40	ppm	LP-AAL-23002
Moisture	-	0.20	Weight %	ASTM E203
Dipropylene glycol	-	0.10	Weight %	LP-AAL-21008

3.2 Toxicity. When evaluated in accordance with 4.2.2, the PGDN shall pose no serious or high risk to the health of personnel or the environment when used for its intended purpose (see 4.2.2 and 6.4).

3.3 Prohibited materials. The PGDN shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2 for NAVSEA programs or NAS411-1 for all other programs.

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3.4 Physical and chemical requirements. The physical and chemical properties of the PGDN shall conform to the limits specified in table II when tested as specified herein. If the lot fails to conform to any of the requirements specified, the lot shall be rejected.

TABLE II. Physical and chemical requirements of PGDN.

Property	Requirements		Test Method
	Minimum	Maximum	
Nitrogen (% by wt)	16.82	16.92	4.2.5
Acidity (as sulfuric acid) (% by wt)	-	0.001	4.2.6
Alkalinity (as sodium carbonate) (% by wt)	-	0.001	4.2.6
Moisture (% by wt)	-	0.50	4.2.7
Stability (potassium iodide at 82 °C) (minutes)	10	-	4.2.8

3.5 Workmanship. The PGDN shall be free from dirt, sediment, and other suspended foreign matter when examined visually by transmitted light (see 4.2.4).

4. VERIFICATION

4.1 Conformance inspection. The material shall be subjected to all the inspections and tests specified in 4.2. When specified (see 6.2), the supplier shall submit a report giving the results obtained for all inspections and tests performed and a certified statement that the lot meets all the requirements of this specification. Unless otherwise specified (see 6.2), all chemicals shall be American Chemical Society (ACS) grade in accordance with A-A-59282, and distilled water shall be used. Where applicable, blank determinations shall be run and corrections applied where significant. Failure of any sample to meet any requirement of this specification shall cause rejection of the lot or batch.

4.1.1 Lot formation. A lot shall consist of the PGDN offered for acceptance at one time that has been produced by one manufacturer, at one plant, during a period of time not exceeding 20 hours, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.6).

4.1.2 Sampling. A sample of approximately 50 milliliters (mL) of PGDN shall be removed from each lot. The sample shall be transferred to a stoppered bottle made of rubber or an approved plastic. The bottle shall be labeled to show the lot number and date of manufacture.

4.1.2.1 Sample bottle cleaning. After disposal of the PGDN from the sample bottle, the bottle shall be cleaned as follows:

a. Rinse thoroughly with acetone.

b. Flush generously with hot tap water (3 minutes under a direct flow is considered adequate).

c. Rinse with distilled water and dry.

4.2 Methods of inspection.

4.2.1 Material. The propylene glycol used for manufacture shall be in accordance with table I. When specified (see 6.2), a list of performed tests and results shall be certified by the manufacturer.

4.2.2 Toxicity. A Health Hazard Assessment (HHA) will be conducted to ensure conformance to 3.2 as specified (see 6.2). The Navy and Marine Corps Force Health Protection Command (NMCFHPC) will evaluate the PGDN using data provided by the manufacturer/distributor to the NMCPHC (see 3.2 and 6.4).

4.2.3 Prohibited materials. Prohibited materials shall be verified as specified (see 3.3 and 6.2).

4.2.4 Visual inspection. All samples shall be visually inspected to determine conformance to the workmanship requirement of 3.5.

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4.2.5 Determination of nitrogen. The nitrogen content shall be determined as follows:

a. Equipment.

(1) 798 MPT Titrino titrator or equivalent

(2) Analytical balance with accuracy ±0.1 milligram (mg)

(4) Glass and platinum electrodes, titration probes

(5) 250-mL beakers

(6) Stir bars

(7) Dropper bottles

b. Chemicals.

(1) Sulfuric acid (H_2SO_4) [CAS 7664-93-9], 95.0 to 98.0 percent (weight to weight)

(2) Potassium nitrate (KNO_3) [CAS 7757-79-1], primary nitrogen standard/certified or equivalent

(3) Ferrous sulfate heptahydrate [CAS 7782-63-0], crystalline

(4) Deionized (DI) water

c. Standardization.

(1) Prepare ferrous sulfate titration solution, 0.4 percent N/mL, as follows:

(a) Dissolve 350 grams (g) of $FeSO_4 \cdot 7H_2O$ in 800 mL of DI water and cool. Slowly add 1 liter (L) cold 1:1 H_2SO_4/H_2O. Add 200 mL of DI water to dilute to 2 L. Store in a cool dark place.

(2) Transfer approximately 10 g of KNO_3 into a porcelain mortar and grind behind a shield to the consistency of table salt using a pestle.

(3) Transfer the KNO_3 to a weighing dish with a lid. Place the weighing dish with lid in oven at 105 °C for 2 hours. If the KNO_3 is out of the oven for more than 4 hours, re-dry at 105 °C for 2 hours. After drying, remove the weighing dish with lid containing the KNO_3 from the oven and place in a desiccator to cool.

(4) Purge the titrator buret containing the ferrous sulfate solution three to four times before beginning the titration.

(5) Raise the titration electrodes from the clean sulfuric acid (where they are stored) and allow them to drip back into the beaker. Dab the electrodes with laboratory wipes and place the contaminated wipes into a beaker full of tap water.

(6) Clean the titration electrodes with DI water and wipe completely dry.

(7) In a clean, dry 250-mL beaker, add a stir bar and 150 mL of sulfuric acid. Place the beaker in the water bath on the stir plate and turn on stirrer.

(8) Weigh 0.50±0.05 g of dry KNO_3. Slowly pour the KNO_3 into the swirl of acid in the beaker and allow the KNO_3 to fully dissolve.

(9) Turn the circulating bath on (10 to 15 °C) and lower the electrodes and dispensing tip into the acid. Begin the titration and allow the sample to titrate to completion.

(10) Once the endpoint is reached, turn off the stirrer and water bath and record the volume (mL) of ferrous sulfate solution used.

(11) Raise the electrodes and allow excess acid to drip into the beaker.

(12) Place the electrodes back into the beaker of clean sulfuric acid and discard the contents of the sample beaker accordingly. Electrodes are stored in clean sulfuric acid when the system is in use and during shutdown. The dispensing tip should be placed in the dispenser tip holder.

(13) Calculate the nitrogen equivalence factor (F) for the ferrous sulfate, as follows:

$$F = 13.855 \times \frac{W}{V}$$

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Where:

F = factor, percent of N per mL

W = weight of KNO_3, in g

V = volume of titrant, in mL

(14) Perform standardization in duplicate on the first day of the work week. The nitrogen equivalence factor of replicates should agree within 0.0005. Repeat each day and ensure replicates agree within 0.0005.

d. Procedure.

(1) Raise the titration electrodes from the clean sulfuric acid (where they are stored) and allow them to drip back into the storage beaker. Dab the electrodes with laboratory wipes and place the contaminated wipes into a beaker full of tap water. Clean the titration electrodes with DI water and wipe completely dry.

(2) In a clean, dry 250-mL beaker, add a stir bar and then approximately 150 mL of sulfuric acid. Place the beaker on the stir plate and turn on the water bath (10 to 15 °C) and stirrer. Allow the acid to cool prior to adding a sample.

(3) Add 0.2 g weighted by difference (approximately seven drops) of PGDN into the swirl of the acid and wait for the sample to disperse in the acid.

(4) Lower the electrodes and dispensing tip into the acid and start the titration. Allow the sample to titrate to completion.

(5) Once the endpoint is reached, turn off the stirrer and water bath and record the volume (mL) of ferrous sulfate solution used.

(6) Raise the electrodes and allow excess acid to drip into the sample beaker.

(7) Place the electrodes back into the beaker of clean sulfuric acid and discard the contents of the sample beaker accordingly. Electrodes are stored in clean sulfuric acid when the system is in use and during shutdown. The dispensing tip should be placed in the dispenser tip holder.

(8) Calculate the nitrogen content in PGDN, as follows:

$$\%\, N = V \times \frac{F}{W}$$

Where:

V = volume of titrant, in mL

F = equivalence factor, percent of N per mL

W = weight of PGDN, in g

(9) Perform test on each lot in duplicate. Replicates should agree within 0.05 percent nitrogen.

4.2.6 Determination of acidity and alkalinity. The acidity and alkalinity of the PGDN shall be determined as follows:

a. Place approximately 50 mL of distilled water into each of two 125-mL Erlenmeyer flasks and add 10 drops of bromothymol blue indicator to each flask.

b. Neutralize the contents of the flasks with either 0.01N acid or base to bring the water to a neutralization point (yellow pH 6.0 – blue pH 7.6). A pale green color indicates the desired neutralization point.

c. Weigh approximately 5 to 10 g of raw PGDN from a weighing bottle into one of the flasks. Allow the sample to enter along the inner sides of the flask. Record the weight of the sample to the nearest 0.1 mg.

d. Gently mix the PGDN and water with a swirling motion. Note any change in color due to the indicator change. Compare the flask containing the PGDN with the blank flask. If the PGDN flask is blue, titrate with standard acid to the neutral point. If the PGDN flask is yellow, titrate with standard base to the neutral point.

e. Use the color of the blank flask to compare the color for the end point. The desired end point is a pale green color.

f. Calculate the percent acidity and percent alkalinity as follows:

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$$Percent\ acidity\ as\ H_2SO_4 = \frac{(mL\ base\ \times normality\ of\ base) \times 4.9}{Sample\ weight, g}$$

$$Percent\ alkalinity\ as\ Na_2CO_3 = \frac{(mL\ acid \times normality\ of\ acid) \times 5.3}{Sample\ weight, g}$$

4.2.7 Determination of moisture content. The moisture content shall be determined as follows:

a. Add 25 mL of methanol to the titration flask.

b. Titrate the methanol to a potentiometric end point with stabilized Karl Fischer (KF) reagent.

c. Weigh and transfer a 2.0-g sample of PGDN to the titration flask.

d. Titrate the sample to a potentiometric end point.

e. Obtain results from KF titrator or calculate the moisture content as follows:

$$Percent\ moisture\ content = \frac{100(KF)}{W}$$

Where:

K = mL of KF reagent used in titration

F = reagent factor (g of water per mL of reagent)

W = weight of PGDN, in g

4.2.8 Determination of stability. The stability shall be determined as follows:

a. Filter a portion of the PGDN sample through two thicknesses “S. and S.” Number 604 filter paper or equivalent.

b. Transfer a 2-mL portion of the filtered material by means of a pipette to each of three test tubes that are 135 to 140 millimeters (mm) long, with a 13-mm internal diameter and 16-mm outside diameter. Ensure no droplets of PGDN are left on the sides of the test tubes during transfer.

c. Stopper each tube by means of a new, tightly fitting cork through which passes a tightly fitted glass rod equipped with a platinum holder for a strip of standard potassium iodide starch-indicator paper.

d. Using forceps, hang a strip of standard potassium iodide starch-indicator paper on the platinum holder. The standard potassium iodide starch-indicator test paper shall normally be approximately 25 mm long and 9.5 mm wide.

e. Moisten a horizontal section in the lower half of the standard test paper with a 50 percent by volume solution of pure glycerin in distilled water. This is conveniently accomplished by dipping a small diameter glass rod into the glycerin-water solution and making contact with the side of the container as the rod is withdrawn so as to minimize the volume of the solution adhering to the rod. Draw the rod across the paper strip so as to produce a level and distinct line of demarcation on the lower part of the test paper.

f. Prepare a blank by suspending a moistened strip of the standard test paper in a clean dry tube.

g. Adjust the temperature of the heat water bath to 82.2±l °C, and insert all four test tubes into the heating water bath. The depth of immersion of the test tubes into the water bath should be approximately 50 mm. Note the time of insertion of the tubes into the bath.

h. During the test, the line of demarcation on the standard test paper should be 76 mm above the level of the PGDN in the test tube. The line of demarcation in the blank tube is regulated at an equivalent height in the tube.

i. Consider the end point of the test to be the first appearance of any discoloration at the line of demarcation between the wet and the dry portion of the test paper in the sample tube which is in excess of the discoloration observed at the same position on the test paper in the blank. Note the time for completion of the test to the nearest minute.

j. Desensitize the sample in each test tube by diluting with ethanol from a squirt bottle before removing the test tubes from the water bath.

k. Consider the minimum time for the three tubes to represent the heat test value of the sample.

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l. After the test, rinse all rods and tubes with acetone to remove the PGDN. Wash with warm soapy water, rinse thoroughly with tap water, then with distilled water, and dry in a steam oven at approximately 90 °C.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. PGDN is intended for use as a propellant ingredient. Current stock of PGDN may be used until depleted; future requisitions for PGDN should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Whether certification of propylene glycol, including tests performed and results, is required (see 3.1, 4.2.1, and 6.7).

c. Requirements for an HHA (see 3.2 and 4.2.2).

d. Requirement for prohibited materials verification (see 3.3 and 4.2.3).

e. Whether a certified analysis and a test report are required (see 4.1).

f. Grade of chemicals if other than ACS (see 4.1).

g. Packaging requirements (see 5.1).

h. Safety precaution requirements (see 6.3).

6.3 Safety precautions. The safety precaution requirements of DoD Manual 4145.26, DoD Contractor's Safety Manual for Ammunition and Explosives, are applicable and should be specified in the contract or order as required by 32 CFR (see 6.2).

NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of NAVSEA OP 5, Volume 1, Ammunition and Explosives Safety Ashore, should be made applicable.

6.4 Toxicity evaluation. The NMCFHPC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCFHPC to the Government for evaluation. The HHA process is described on the NMCFHPC's website, https://www.med.navy.mil/Navy-Marine-Corps-Public-Health-Center/.

6.5 Toxic effects. Toxic effects may occur from the inhalation of vapors from PGDN or the absorption from direct contact with the skin.

6.6 Batch. A batch is defined as that quantity of PGDN that has been subjected to the same unit chemical or physical process intended to make the final product homogeneous.

6.7 Guidance for propylene glycol. Propylene glycol is used to make PGDN, which is then formulated into a propellant. Gas chromatography analysis showed unacceptable levels of contamination in historical lots of propylene glycol, which deemed the material unfit for nitration. The propylene glycol made to the requirements in table I is of desired purity and limits the known contaminant, dipropylene glycol. Retested propylene glycol will often accumulate moisture over time due to it being highly hydroscopic; 0.5 percent maximum moisture content is acceptable for retested propylene glycol.

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6.8 Subject term (key word) listing.

1,2-Propanediol
PGDN

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – OS
(Project 9135-2023-001)

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